

Temperature vs Top-p

Two ways to control how a language model picks the next token -- explained simply.

The 3-Sentence Explanation

1

Temperature controls how adventurous the model is when picking the next word -- low temperature makes it play it safe and almost always pick the most likely word, while high temperature makes it spread its bets and sometimes pick surprising words.

2

Top-p works differently: instead of reshaping probabilities, it draws a cutoff line -- it adds up the most likely tokens one by one until their combined probability reaches p (e.g. 90%), then bans every other token from being picked at all.

3

The key difference is what each one changes: temperature squeezes or stretches the entire probability bar chart (all tokens get higher or lower bars), while top-p simply decides which tokens are allowed in the running and which are not.

TEMPERATURE

Analogy: Volume knob

Turns all voices up or down equally.
Low volume -> only the loudest voice heard clearly (greedy pick).
High volume -> many voices audible
(more random selection).

VS

TOP-P

Analogy: Guest list

Only invites tokens whose cumulative probability fills the ' p ' budget.
Remaining tokens are simply banned from being picked, no matter what.
Probabilities of eligible tokens re-scaled.

How They Work Together (most APIs apply both)

Step 1 Temperature

Reshape the full distribution
(spikier or flatter)

Step 2 Top-p

Cut off tokens below the cumulative probability threshold

Step 3 Sample

Pick one token from the surviving eligible set

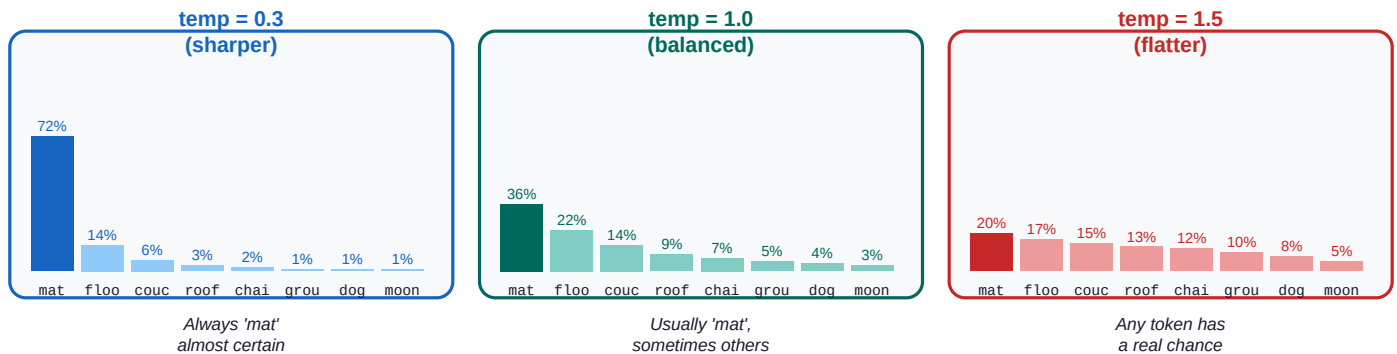
Visualising the Difference

Same starting prompt: "The cat sat on the ____" Same model. Same raw probabilities.

How Temperature Reshapes the Whole Distribution

Imagine the model produced these raw probabilities for the next token. Temperature multiplies or divides every bar. It does not remove any token -- it changes their relative heights.

Temperature: reshapes the **WHOLE** distribution



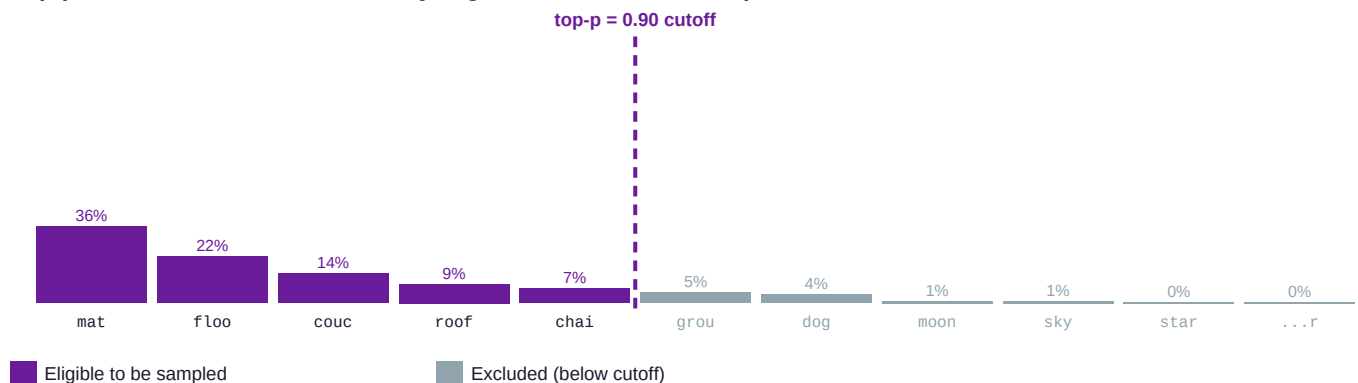
Prompt: "The cat sat on the ____" All ~100k tokens get a score; only top 8 shown.

Simple way to remember: think of temperature as a dimmer switch on a room full of voices. Turn it down (low temp) and only the loudest voice is audible -- the model always picks that one. Turn it up (high temp) and quieter voices become loud enough to hear -- the model might pick any of them.

How Top-p Draws a Cutoff Line

Top-p does not reshape the bars. It walks down the list from most-probable to least-probable and keeps adding tokens to the 'eligible' list until their combined (cumulative) probability reaches p. Every token after that is simply banned. The remaining eligible tokens are then re-scaled to sum to 100% and one is sampled.

Top-p: draws a **CUTOFF LINE** — only eligible tokens can be sampled



Simple way to remember: think of top-p as a guest list with a capacity limit. The venue (probability budget) holds exactly p% worth of guests. The most popular tokens fill seats first. Once the venue hits capacity, the door closes -- no matter how famous the remaining tokens are, they cannot get in. top-p = 0.9 means the venue fills to 90% capacity before the door closes.

Concrete Examples with Numbers

The same scenario, showing exactly what each parameter does step by step.

Example A: Temperature in action

Prompt: "The weather today is ____" Raw model probabilities before temperature is applied:

Token	Raw prob	After temp=0.3 (sharper)	After temp=1.0 (unchanged)	After temp=1.8 (flatter)	Eligible at all?
"sunny"	45%	81%	45%	27%	Yes
"cloudy"	25%	13%	25%	22%	Yes
"warm"	15%	4%	15%	18%	Yes
"perfect"	8%	1%	8%	16%	Yes
"bizarre"	4%	~0%	4%	11%	Yes
"apocalyptic"	3%	~0%	3%	6%	Yes

At temp=0.3: "sunny" dominates at 81% -- almost always picked. At temp=1.8: "apocalyptic" rises from 3% to 6% -- now a real possibility.

Example B: Top-p in action

Same prompt, same raw probabilities. Showing which tokens survive different top-p values:

Token	Raw prob	Cumulative	top-p=0.5 eligible?	top-p=0.8 eligible?	top-p=1.0 eligible?
"sunny"	45%	45%	Yes	Yes	Yes
"cloudy"	25%	70%	No (70% > 50%)	Yes	Yes
"warm"	15%	85%	No	No (85% > 80%)	Yes
"perfect"	8%	93%	No	No	Yes
"bizarre"	4%	97%	No	No	Yes
"apocalyptic"	3%	100%	No	No	Yes

top-p=0.5: only "sunny" is eligible (cumulative 45% still under 50%; adding "cloudy" would push to 70% -- over budget, so "cloudy" is excluded).
top-p=1.0: everything is eligible -- no filtering at all.

Code Examples + When to Use Each

Python examples using the OpenAI API, plus a quick-reference guide.

Setting Temperature and Top-p in Python

```
import openai

client = openai.OpenAI()

def generate(prompt: str, temperature: float, top_p: float) -> str:
    response = client.chat.completions.create(
        model="gpt-4o-mini",
        messages=[{'role': 'user', 'content': prompt}],
        temperature=temperature,
        top_p=top_p,
        max_tokens=150,
    )
    return response.choices[0].message.content

prompt = "The weather today is"

# ---- Scenario 1: precise / factual output ----
# Low temperature + high top-p = confident, consistent, no surprises
print(generate(prompt, temperature=0.2, top_p=1.0))
# -> 'The weather today is sunny and warm.' (same every run)

# ---- Scenario 2: creative writing ----
# Higher temperature + slightly restricted top-p = varied but not nonsensical
print(generate(prompt, temperature=1.0, top_p=0.9))
# -> 'The weather today is perfect for a long walk.' (run 1)
# -> 'The weather today is cloudy but oddly beautiful.' (run 2)
# -> 'The weather today is warm and restless.' (run 3)

# ---- Scenario 3: maximum diversity (brainstorming) ----
# High temperature + high top-p = full exploration of possibilities
print(generate(prompt, temperature=1.5, top_p=1.0))
# -> 'The weather today is hauntingly still.' (unexpected but valid)
# -> 'The weather today is bizarre, almost apocalyptic.' (rare token picked)

# ---- Scenario 4: guard rails only (top-p does the safety work) ----
# temp=1 keeps raw probs, top-p=0.5 bans all low-prob tokens
print(generate(prompt, temperature=1.0, top_p=0.5))
# Only 'sunny' (45%) qualifies under top-p=0.5 cutoff
```

```
# -> 'The weather today is sunny.' (varied phrasing but token is fixed)

# ---- Important: OpenAI recommends changing ONE at a time ----

# Do NOT set both temperature and top_p to non-default values simultaneously

# unless you have a specific reason to. Pick one to tune.
```

Temperature vs Top-p: Side-by-Side Comparison

Dimension	Temperature	Top-p (nucleus sampling)
What it changes	The height of every bar in the probability chart	Which tokens are allowed to be picked at all
Default value	1.0 (raw distribution, no change)	1.0 (all tokens eligible, no cutoff)
Value = 0 / 0.0	Greedy: always pick the single highest-probability token	Only 1 token eligible (the most probable one)
High value	Flat distribution -- any token can be picked	More tokens allowed in -- wider variety
Low value	Peaked distribution -- top token dominates	Fewer tokens allowed in -- tighter, safer output
Scale	0.0 to 2.0 (above 1.5 is rarely useful)	0.0 to 1.0 (above 0.95 adds little extra variety)
Best for	Controlling overall 'boldness' of the model's choices	Cutting out nonsense / very unlikely tokens as a safety net
Analogy	Dimmer switch: changes brightness of all voices equally	Guest list: door closes once venue reaches p% capacity

Key Takeaways

— Temperature reshapes the full distribution -- every token's probability changes
— Top-p draws a line and excludes everything below the cumulative threshold
— Both are applied to the same raw probabilities, before any sampling happens
— OpenAI recommends tuning only one at a time -- change temperature OR top-p, not both
— temp=0 is greedy (always picks the top token). temp=1 samples the raw distribution
— top-p=1.0 means no filtering (all tokens eligible). top-p=0.5 keeps only the top tokens that fill 50% of probability